

National Reference Metadata in ESS Standard for Quality Reports Structure (ESQRS)

Country: Greece

Compiling agency: Hellenic Statistical Authority (ELSTAT)

Domain name: Structure of Earnings Survey 2022

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2. Statistical presentation

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2.1 Data description

The Structure of Earnings Survey 2022 (SES 2022) is the sixth of the series of four-yearly surveys carried out by the Hellenic Statistical Authority (ELSTAT) under the Council Regulation (EC) No 530/1999, the Commission Regulation (EC) No 1916/2000 (amended) and the Commission Regulation (EC) No 1738/2005.

This quality report is prepared in accordance with the Commission Regulation No 698/2006 of 5 May 2006 implementing Council Regulation (EC) No. 530/1999 concerning quality evaluation of structural statistics on labour costs and earnings.

The main purpose of the survey at national level is to produce statistics on the structure of earnings of employees. More specifically, the collected variables refer to:

- the enterprise (branch of economic activity, geographical region, number of local units, number of employed persons, type of employment contract, form of economic and financial control) and
- the employees (sex, age, occupation, citizenship, status in employment, length of service in the enterprises, educational attainment, earnings during the year of the survey, hourly earnings, contributions, paid hours, etc.)

From a microeconomic perspective, these statistics are a tool for drawing conclusions on the structure of earnings in the branches of economic activity. Moreover, the allocation of earnings may be assessed and interpreted on the basis of the characteristics of the persons employed in the different branches of economic activity.

At European level, the survey aims at compiling harmonized statistics providing comparable quantitative and qualitative information for all the EU Member States pertaining to the structure of earnings of employees.

2.2 Classification system

Economic activity

Economic activity is coded in Nace Rev.2.

More specifically:

- For the years 2002 and 2006 the data were compiled in accordance with Nace Rev.1.1
- In 2010, the survey was conducted in line with Nace Rev.2, at the same time, the data were also coded in accordance with Nace Rev.1.1, in order to ensure comparability with the previous surveys (2002 and 2006).
- In 2014, 2018 and 2022, the survey was conducted in line with Nace Rev.2

Occupation

All occupations are covered, excluding Major group 6 “Skilled agricultural, forestry and fishery workers” of ISCO_08.

More specifically:

- For the years 2002 and 2006 the data were compiled in accordance with ISCO_88
- In 2010, the survey was conducted in line with ISCO_08, at the same time, the data were also coded in accordance with ISCO_88, in order to ensure comparability with the previous surveys (2002 and 2006).
- In 2014, 2018 and 2022, the survey was conducted in line with ISCO_08

Education

For the years 2002, 2006 and 2010, all education levels are covered and the data were compiled in accordance with the statistical classification ISCED_97. For the years 2014, 2018 and 2022 all education levels are covered and the data were compiled in accordance with the statistical classification ISCED_2011.

Geographical location

Regional breakdown is based on the Nomenclature of Territorial Units for Statistics (NUTS, level 1).

2.3 Coverage - sector

The 2002 survey covered sections C-K of Nace rev.1.1.

The 2006 survey covered sections C-K, M, N and O of Nace rev.1.1.

The 2010 survey covered all the sections B-S (excluding O) of Nace rev.2, at the same time, the data were also coded in accordance with Nace rev.1.1, in order to ensure comparability with the previous surveys (2002 and 2006).

The 2014, 2018 and 2022 surveys covered sections B-S (excluding O) of Nace rev.2.

2.4 Statistical concepts and definitions

The concepts and definitions of the basic variables used for the compilation of data of the structure of earnings survey are laid down in Regulation (EC) 1738/2005 amending Regulation (EC) No 1916/2000 and Regulation (EC) 530/1999 of the European Parliament and of the Council.

Regional Unit is a geographical area and it is defined in accordance with the Municipalities under Kallikratis Plan. In most cases, it refers to a geographical area which is defined by the boundaries of the Department (Nomos).

Local unit of the enterprise is every local branch of the enterprise which is operating in a different physical location, it may be classified in a different branch of economic activity and it is located in another Regional Unit.

Economic activity of the local unit is the branch of economic activity in which is classified the local unit and may be different from the branch of economic activity of the enterprise.

Group of enterprises

An enterprise group is an association of at least two enterprises bound together by legal and/or financial links. A group of enterprises consists of the parent enterprise and the affiliated companies.

Collective pay agreement is any type of agreement signed between the enterprise and the employees or their collective body and may be:

- A national level agreement
- An industry agreement
- An agreement for individual industries in individual regions where the enterprise or the local unit are located
- An enterprise agreement or an agreement applying only to the workers of a local unit.

Employed persons, for the purposes of the survey on the structure of earnings, are all the employees of the enterprise, i.e., the salaried personnel and the wage earners.

Employees are the employed persons who work and receive remuneration in the form of wages or salaries and have a direct employment contract with the enterprise and are remunerated on the basis of the number of hours worked (full-time or part-time) and the duration of the contract (fixed or indefinite).

The following categories are excluded:

- management personnel whose remuneration wholly takes the form of a share in profits,
- unpaid family workers
- outworkers that are remunerated on the basis of the product or service they deliver, sales representatives who are not included in the payroll, that is they are self-employed
- occasional workers (i.e. employed without pre-defined working hours),
- interim workers (employed by/through agencies),
- persons wholly remunerated by way of fees, or commission,
- members of the board of directors of the enterprise,
- unpaid owners, directors or managers (paid by way of profit share or by fee),
- workers of the observation unit permanently (for more than one year) working in an affiliated company located abroad,
- voluntary workers.

Full-time employees are the members of the personnel whose regular working hours are the same as the collectively agreed or customary hours worked in the enterprise, even if their contract is for less than one year.

Part-time employees are the members of the personnel whose regular working hours are less than the collectively agreed or customary hours worked in the enterprise on a daily, weekly or monthly basis.

Seasonal employees are the members of the personnel that are hired under a special work contract designed to cover seasonal needs of the enterprise or the local unit, which are identified in the same time period every year.

Apprentices are those remunerated employed persons who do not fully participate in the production process and work either under an apprentice contract or in the context of a project for which vocational training is most important than productivity.

Occupation, according to the statistical classification of occupation ISCO_08 is the “real” worked mainly performed by the employee. The essential information for determining the occupation is usually the employee’s job title and a description of the main tasks undertaken in the course of his/her duties.

Paid hours cover normal and overtime remunerated hours, hours remunerated with a lower rate even if the difference is covered by social security contributions, as well as hours not worked during the reference periods but nevertheless paid (annual leave, sick leave, official holidays and other hours paid such as for medical examinations).

Special earnings are the payment for shift work, night work and work during weekends, given the fact that they are not counted as overtime pay.

Payments in kind are all the goods and services provided to the employees by the enterprise or the local unit, such as products of the enterprises provided for free or at a reduced price, (e.g. food, beverages, footwear, clothes) shelter for the personnel, leased cars, as well as share of the company and other benefits.

Total annual bonuses refer to any irregular payments which did not occur during each pay period, such as holiday bonuses, Christmas and Easter bonuses (13th and 14th salary), productivity bonuses, occasional commissions, profit-shares, etc. Bonuses paid regularly on a monthly basis are excluded.

2.5 Statistical unit

The primary statistical unit is the local unit of the enterprise with average annual employment amounting to, at least, 10 persons and the final statistical unit is the employee. As a register of local units is not available, the primary unit is considered the enterprise with average annual employment amounting to, at least, 10 persons, classified in sections B-S (excluding O) of Nace Rev.2.

2.6 Statistical population

The statistical population is the total of the enterprises employing at least 10 employees and classified in sections B-S (excluding O) of Nace Rev.2.

2.7 Reference area

Greece, total.

2.8 Coverage-Time

2022

2.9 Base period

Not applicable.

3. Statistical processing

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3.1 Source data

The Structural Earnings survey covered all local units belonging to enterprises with average annual employment 10 persons or more in the areas of economic activity defined by sections B-N plus P, Q, R and S of NACE Rev.2. The sampling frame that was used for the selection of the primary sampling units (enterprises) was the updated Business Register with reference year 2020.

In order to trace the local units for data collection, since the survey units are the local units and those were not included in the register of the enterprises, we applied a two-stage stratified sampling. At the first stage, a sample of n_h enterprises out of the N_h enterprises was selected in each stratum h with equal probabilities, and next in each selected enterprise a sample of local units was selected.

The following stratification criteria for the enterprises were applied:

- Geographical Region – NUTS 1 (Northern Greece, Central Greece, Attica, Aegean Islands and Crete)
- Two-digit NACE Rev.2 code of economic activity within each geographical region
- Size class of the enterprise. In each of the major strata, the enterprises were stratified into 10 size classes, according to their size, determined by their average annual number of employees in the business register, as follows.

Size class	Number of employees
1	10-19
2	20-49
3	50-99
4	100-249
5	250-499
6	500-999
7	1000-1999
8	2000-4999
9	5000-9999
10	10000+

Size classes 5 and over (250+ employees) were exhaustively surveyed (census classes).

The total number of employees of the local unit was not included in any register but this information was collected and enumerated by the interviewer for each selected enterprise of the sample. A sample of local units was selected from the local units that belong to the enterprises of the sample which were selected at the first stage. Subsequently, m_{hi} employees were selected out of the M_{hi} employees belonging in the local units during the survey period ($=1, 2, \dots, n_{hi}$) with equal probabilities by applying systematic sampling. The sample size of employees in local units was based on the proportional allocation of the employees of the enterprise, according to the size of the local units.

The final sample size of legal units was 6,871 out of 46,753 (sampling rate=14.7%) and the final sample size of employees was 551,927 out of 2,120,574 (sampling rate=26%).

3.2 Frequency of data collection

The survey on the structure of earnings is conducted every four years.

3.3 Data collection

For the compilation of SES 2022 multiple administrative and other sources were used, while a small number of data were collected directly by enterprises through a specially designed questionnaire (questionnaire form 1 and form 2).

The questionnaire Form 1 is the questionnaire used for the enterprise and it helps to collect information on the enterprise, such as economic activity, number of persons employed, the form of economic and financial control, number of local units of the enterprise, main type of employment contract and whether the company belongs to a Group of enterprises.

The questionnaire Form 2 is used to collect information on the employees. More specifically, the collected data pertain to sex, age, educational attainment, occupation, position, employment status, the total length of service in the enterprise, full or part-time employment. In addition, financial data are collected for October of the year when the survey is conducted, such as data on hours paid, overtime pay, on gross earnings, payments related to overtime, and payments related to shift work. Furthermore, annual financial data are collected, such as gross earnings, payments in kind, and payments related to annual bonuses.

Both questionnaires were developed on paper and in electronic format as a user-friendly editable pdf.

The questionnaires are delivered to external survey workers (statistical interviewers) who, once trained, visit the enterprises in order to fill in the questionnaire form 1 with the assistance of the enterprise's staff or they hand the questionnaire to be filled in by the accountant of the enterprise. Then, they collect the filled in questionnaires and deliver them back to ELSTAT. Before filling in the questionnaire form 2, they have to be supplied with the payroll list for October (of the specific reference year), which will be used as sampling frame in order to draw the sample of the employees to be surveyed. Then, in cooperation with the staff of the enterprise, they have to fill in the questionnaire form 2.

In cases where the enterprise refuses to cooperate with the statistical interviewers or refuses to respond, or delays or in case there are particularities, an employee from the Collection of Business Survey Data Section of ELSTAT pays a visit to the enterprise and collects the necessary information himself. In some cases (due to lack of time) the questionnaire may be delivered and collected by e-mail.

The same procedure is followed in the Regional Statistical Offices (RSO) of ELSTAT. The questionnaires are sent by post to the RSO and then the completed questionnaires are sent by the RSO back to ELSTAT headquarters by post. In some cases (due to lack of time) the questionnaire may be sent and returned by e-mail

Non-response is addressed through telephone contact with the enterprise, reminders sent by e-mail, or personal visits to the enterprise.

For all the enterprises of the sample and for all of their employees (and not a sample of them), data were used from the administrative and other sources. In cases of many erroneous or missing data, the respective enterprises and employees were excluded from the final results. Readjusted weighting factors were used for the calculation of the final SES results.

3.4 Data validation

Global checks and Plausibility checks have been applied on micro data based on Eurostat's checks listed in document "Eurostat's arrangements for implementing the Council Regulation 530/1999, the Commission Regulations 1916/2000 and 1738/2005".

The data are validated by means of a series of logical checks to ensure that the data are reasonable and consistent with other variables. During data processing, the data are checked in order to identify and duly correct any errors. The checks done on administrative data, was the coverage of the source in terms of employers and employees and the consistency of its fields to SES variables. The coverage and the consistency were satisfactory. Comparisons at aggregate level are made, mostly as regards employment data, with the data of the Labour Force Survey and as regards average monthly salaries with National Accounts. Then the data are checked by employment stratum and by 2-digit branch of economic activity. In case big differences are observed, all the enterprises belonging to the same stratum are investigated. Once identified, the errors are further investigated in cooperation with the enterprise in order to confirm that it is an error or it is just about an unusual value.

3.5 Data compilation

Please see attached file.

Annexes:

[Estimation process](#)

3.6 Adjustment

Not applicable.

4. Quality management

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4.1 Quality assurance

Correctness and completeness checks, as well as quality checks, are carried out during the whole process of the compilation of the data of the survey: from the data collection stage to the final compilation of the results.

First of all, well-trained and experienced staff is utilized for all the stages of the compilation of the data, namely for data collection - including communication with the enterprise - initial checks, data entry, checks for outliers, and final checks concerning the time series characteristics of every enterprise.

Data are validated either before or after data entry by means of logical checks. During data processing, the data are checked in order to identify and correct any errors. When an error is identified, the data are further examined, in cooperation with the enterprises, in order to confirm that it is an error or it is just an unusual value. At the same time, data are checked for completeness, accuracy, and consistency of the correlating variables.

The Hellenic Statistical Authority (ELSTAT) aims to ensure and further improve the quality of statistics produced and maintain the confidence of users in them. This is achieved through the [Quality Policy](#) and [Quality Guidelines](#) of ELSTAT, which are posted on the website of ELSTAT.

4.2 Quality management-assessment

The results of the survey on the structure of earnings are reliable, they are transmitted to Eurostat where they are assessed and validated.

5. Relevance

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5.1 Relevance - User Needs

The survey on the structure of earnings meets national needs and the needs of European users.

Generally, the survey provides statistical information necessary to improve the competitiveness and performance of the business community.

The main users of the survey on the structure of earnings are:

- International Organizations (Eurostat, OECD, European Central Bank (ECB), International Monetary Fund (IMF), International Labour Organisation (ILO))
- National authorities (Government, Ministries),
- Services and Institutions (Banks, Universities, Research Institutes, etc.)
- Social Partners and Trade Unions
- The media, researchers, students, and the general public

Moreover, the enterprises that have the obligation to fill in the questionnaire are at the same time users of these data. They request relevant data in order to design their remuneration policy, estimate the costs of contractual agreements, and study competitiveness in the branch of economic activity they belong to.

5.2 Relevance - User Satisfaction

A proxy of Relevance can be provided with the help of the user satisfaction survey. ELSTAT conducts:

a. User Satisfaction Survey

The [User Satisfaction Survey](#) is conducted by ELSTAT on a daily basis, through a questionnaire that should be filled in by users each time they request and are provided with data by ELSTAT. The Statistical Data Dissemination Section and the Library of ELSTAT, drawing information from the User Satisfaction Survey compile an annual report presenting data on the number of users, the responsiveness level to users' requests, the kind of the requested data, as well as the dissemination mode of the statistical information.

b. Users' Conference

Pursuant to its annual statistical program, ELSTAT has been conducting since 2010, on a yearly basis, the [Users' Conference](#) with the participation of representatives of the public and the private sector, academics, and research institutes.

The Users' Conferences offer ELSTAT the opportunity to gather comments and proposals made by users of statistics regarding the dissemination of statistical information, access to statistical data, and any shortcomings in the produced statistics.

The Users' Conferences help ELSTAT to derive useful conclusions on the areas of statistical production that need to be improved in order to better fulfill the changing needs of users. Such conclusions are incorporated in the annual and short-term statistical work programs of ELSTAT.

More information on a) the results of the User Satisfaction Survey and b) on the Users Conferences are available on the portal of ELSTAT.

Based on users' Statistical data requests, the most wanted SES data are earnings breakdowns by occupation, gender, level of education, working time conditions, the existence of collective pay agreement, economic activity, and different combinations between these variables.

5.3 Completeness

The 2022 SES satisfies the needs of the main user – European Commission (Eurostat) in terms of:

- Coverage: enterprises with 10+ employees in NACE Rev. 2 sections B - S, excluding O.
- Completeness: The survey covers all (100%) mandatory variables and optional variables

5.3.1 Data completeness - rate

Not applicable.

6. Accuracy and reliability

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6.1 Accuracy - overall

The errors that have an impact on accuracy of the results are: (a) sampling errors and (b) non-sampling errors.

(a) Sampling errors are due to the fact that the survey is not conducted on all the enterprises of the target population but on a sample of these enterprises. The sample size ensures high accuracy estimation at the level of Greece total.

(b) Non-sampling errors mostly refer to measurement errors and non-response errors, on account of non-response of the sample enterprises. Any measurement errors during data collection are identified by means of quality checks are duly corrected. As regards non-response errors, all efforts are made for contacting the enterprises that have not responded by phone or by other means of communication, in order to persuade them to cooperate and give the required information.

As a consequence, the overall accuracy of the results of the survey on the structure of earnings is generally considered to be high.

6.2 Sampling error

The survey on the structure of earnings is a sample survey and therefore sampling errors are observed in the total of estimations and in the estimations at the level of sections of economic activities.

6.2.1 Sampling error - indicators

The relative sampling error in form of coefficient of variations (CV) of the estimates of Average Gross Hourly Earnings and Gross earnings in the reference month (October 2022) for the year 2022 were calculated with SPSS with clusters being the local units within the enterprise and strata were formed by: NACE Rev.2 at two digit level of the local unit x NUTS1 of the local unit x Size class of the local unit. The CV tables are included in the annex.

6.3 Non-sampling error

Non-sampling errors are basically of 4 types:

Coverage errors: errors due to divergences existing between the target population and the sampling frame.

Measurement errors: errors that occur at the time of data collection. There are a number of sources for these errors such as the survey instrument of collection, the information system.

Processing errors: errors in post-data-collection processes such as data entry, keying, editing and weighting.

Non-response errors: errors due to an unsuccessful attempt to obtain the desired information from an eligible unit.

6.3.1 Coverage error

As the Business Register of ELSTAT on the basis of which the survey was designed, is not fully updated from more recent administrative sources, there were problems of over-coverage and under-coverage.

The over-coverage problems are due to non-active enterprises (closed, in bankruptcy/insolvency, not located) and out of scope enterprises (with less than 10 employees). These enterprises were included in the business register, they were selected in the sample, but they were not actually existed at the time of the survey.

These enterprises actually reduced the initial sample size of enterprises and consequently of local units. The decrease in the number of sampling units in each stratum inflates the variance of the estimated statistics.

The under-coverage refers to the situation when newly emerged or units with renewed activity with 10+ or more employees were not included in the initial sampling frame.

As a result, the under-coverage problem underestimates the produced statistics. Corrections and weighting for non-coverage is difficult, because under-coverage rates cannot be obtained from the sample itself, but only from external sources. The under-coverage was not quantified.

Moreover, due to miss-classification problems of the register, some sampling units changed design strata after data collection. These units were allocated to new strata (employment size class, division of economic activity) maintaining their initial probability of selection. This fact changed the initial variation of data, the initial allocation of the sample enterprises, thus inflating the variation of estimations. Therefore, the coefficient of variation of the produced statistics was higher than the coefficient of variation which had been calculated on the basis of the sample design.

6.3.1.1 Over-coverage – rate

The percentage (%) of enterprises included in the sample but ceased to be active or are in fact out of the scope during data collection period is presented in the following table.

NACE_1dgt	Over-coverage rate (%)
B	0,10
C	0,05
D	0,29
E	0,15
F	0,13
G	0,05
H	0,19
I	0,12
J	0,08
K	0,13
L	0,15
M	0,12
N	0,11
P	0,34

	Q	0,05	
	R	0,18	
	S	0,09	
	Total	0,11	

6.3.1.2 Common units - proportion

Not applicable.

6.3.2 Measurement error

The questionnaires are filled in through personal interviews in the enterprise. The method which is used ensures high-quality data since the statistical interviewers assist the respondents to fill in the questionnaire and /or they fill in themselves the data on employees (questionnaire form 2) and they check very carefully the filled in questionnaires before leaving the enterprise.

The statistical interviewers are normally external survey workers and, if necessary, experienced permanent employees of ELSTAT. The interviewers are duly trained during seminars before the conduct of the survey. The purpose of the training seminars is to enable the interviewers to

- a) fully understand the definitions and the variables of the survey,
- b) correctly fill in the survey questionnaire and
- c) check the questionnaires for any errors through logical checks.

The structure and the size of the questionnaire make it friendly to statistical interviewers and respondents. The questions are clearly stated using the appropriate language and terminology. Furthermore, the compiled guidelines cover analytically all the survey variables.

The support and supervision of the collection of questionnaires were decentralized in the Regional Statistical Offices (RSO) of ELSTAT. The checking for the detection of measurement errors was carried out in the RSO. The RSO were also provided by the central office of ELSTAT with written and telephone guidance on how to process data and deal with arising problems.

The main problem for interviewers is that respondents who keep data differently, do not make further efforts to make them suitable for statistical requirements, or do not understand or read the explanatory notes.

After receiving the questionnaires, the statistical office recorded them, and separate valid questionnaires from those to be revised. If they are not thoroughly completed and corrected, the questionnaire is returned to the interviewer in order to contact the enterprise again, otherwise, it cannot be considered valid.

There is a debugging procedure that assures not missing data, coherence among individual characteristics (age, length of service, level of education, etc), and coherence among economic data (monthly and annual earnings related to the hours paid etc).

If extreme values in data are detected during this phase, a telephone call is made to the respondent for clarifications.

All the above ensure the fact that the collected questionnaires are complete and correctly filled in, thus eliminating any measurement errors.

6.3.3 Non response error

In order to improve the response rate of the survey, the enterprises are addressed by sending reminder letters via fax or email, by calling the enterprises or by paying a visit to the enterprise, in order to collect the requested data. Population weighting adjustment, in the form of poststratification, was applied to reduce bias created by non-response. However nonresponse results in higher standard errors.

6.3.3.1 Unit response - rate

The response rate (%) of enterprises by section is presented in the following table.

Section	Response rate (%)
B	85.4
C	90.3
D	62.7
E	86.9
F	66.1
G	87.4
H	78.9
I	62.3

	J	83.4	
	K	78.6	
	L	69.4	
	M	76.7	
	N	77.9	
	P	47.7	
	Q	70.3	
	R	52.2	
	S	56.0	
	Total	75.6	
6.3.3.2 Item non-response rate			
Not applicable.			
6.3.4 Processing error			
Once the data are gathered in the headquarters of ELSTAT, the following statistical actions take place: • Codification, • Data entry • Data validation (checks of completeness and plausibility checks) • Logical checks • Quality checks are carried out on the collected data in order to identify any extreme values. Any outliers and errors are identified and duly corrected. • Comparison with other sources of statistical information as with the corresponding data of the former relevant survey (reference year 2018) and all possible high differences were investigated. • After the tabulation, the results obtained were analysed in order to know whether they were coherent with the available short-term statistics on labour and wage costs. These activities were carried out by expert, permanent staff of the Section in charge of the survey.			
6.3.4.1 Imputation - rate			
Not applicable			
6.3.5 Model assumption error			
• The monthly questions have as reference October 2022 in the questionnaire because it is a month without seasonal payments and absences. • The accounting or fiscal year coincides with the calendar.			
6.4 Seasonal adjustment			
Not applicable.			
6.5 Data revision - policy			
The data are not subject to revisions after their publication, since they are final and are not based on estimates. Revisions may take place only if there are changes in methodology or the statistical classifications of the branches of economic activity or occupations. More information on the revision policy implemented by ELSTAT is available on the website of ELSTAT.			
6.6 Data revision - practice			
In case the data of the survey on the structure of earnings are revised, this is announced with a relevant News Release.			
6.6.1 Data revision - average size			
Not applicable.			

7. Timeliness and punctuality	Top
7.1 Timeliness	
The final results of the 2022 SES transmitted to Eurostat in June 2024, thus the time lag between the transmission date and the reference period of the data is t+18 months, in the relevant deadline posed in the Regulation.	
7.1.1 Time lag – first result	
Not applicable.	
7.1.2 Time lag – final result	
Not applicable.	
7.2 Punctuality	
All deadlines are met.	
7.2.1 Punctuality - delivery and publication	
The Announcement was published as scheduled.	
8. Coherence and Comparability	Top
8.1 Comparability – geographical	
The collected survey variables are fully in line with the relevant definitions laid down in Regulation (EC) 530/1999 of the European Parliament and of the Council, thus ensuring comparability of statistics among Member States.	
8.1.1 Asymmetry for mirror flow statistics - coefficient	
Not applicable.	
8.2 Comparability over time	
There are no significant differences in the definitions used in the surveys conducted in 2002, 2006, 2010, 2014, 2018 and 2022 apart from those provided by the relevant Regulations, such as the change in the statistical classifications of economic activities and occupations. The 2002 and 2006 surveys were conducted in accordance with Nace Rev.1.1 and ISCO_88. The 2010, 2014, 2018 and 2022 surveys were conducted with Nace Rev.2. and ISCO_08, yet, in order to ensure comparability with the surveys of 2002 and 2006, the 2010 survey was conducted in accordance with Nace Rev.1.1 and ISCO_88 as well.	
All the above surveys are in line with European principles.	
8.2.1 Length of comparable time series	
Not applicable.	
8.3 Coherence - cross domain	
The data of the survey were compared with the data of the Labour Force Survey in terms of employment and hours worked data.	
The comparisons have been made between Gross monthly earnings-SES and Net monthly earnings-LFS for each Nace Rev.2 Section B_S_X_O, with the breakdown Men- Women. The differences observed were above 30% for SES data.	
Annexes: Comparison between SES and LFS	
8.4 Coherence - sub annual and annual statistics	
No comparisons, the survey is conducted only every 4 years.	
8.5 Coherence - National Accounts	
The survey on the structure of earnings covers only the enterprises employing at least 10 persons, while National Accounts cover all the enterprises.	
In order to achieve a level of comparability between National Accounts data and the results of the survey on the structure of earnings, and in order to check their coherence, the earnings per employee must be calculated for each one of these two data sources. When compared, the data prove to be coherent in the majority of the branches of economic activity.	

The comparisons have been made between the number of employees and earnings per employee for each Nace Rev.2 Section B_S_X_O. In most sections as well as the total of Nace rev.2 sections, the value of the number of employees was higher in the National Accounts data. As regards earnings per employee, SES data recorded higher values in most sections.

Annexes: [Comparison between SES and National Accounts](#)

8.6 Coherence – internal

All data are checked for internal coherence by ELSTAT and Eurostat.

9. Accessibility and clarity

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9.1 Dissemination format - News release

An Announcement was published on December 20, 2024.

9.2 Dissemination format – Publications

There are no publications.

9.3 Dissemination format – online database

There is no online database for the results of the Structure of Earnings Survey 2022. Data files are produced on ad hoc basis upon users' requests.

9.3.1 Data tables – consultations

Not applicable.

9.4 Dissemination format - microdata access

Microdata, if not confidential, are made available to users after their submitting a request to the:

Statistical Data Dissemination Section

46, Peiraios and Eponiton Str.,

18510 Piraeus

Tel: +30 213 135 2022

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For confidential reasons, users can have access to microdata, only under a confidentiality contract and with respect to the valid process.

9.5 Dissemination format - other

Not applicable.

9.6 Documentation on methodology

The methodology concerning the compilation of data on Structure of Earnings Survey is laid down in Regulation (EC) No 530/1999 and (EC) 1738/2005 of the European Parliament and of the Council.

The survey is a sample survey using the two-stage stratified sampling method. The primary statistical unit is the local unit of the enterprise with average annual employment amounting to at least 10 persons and the final statistical unit is the employee. As a register of local units is not available, the primary unit is considered the enterprise with average annual employment amounting to at least 10 persons and classified in sections B-S (excluding O) of Nace Rev.2.

9.7 Quality management - documentation

Quality is ensured by ELSTAT by means of logical checks and specialized validation checks by Eurostat.

9.7.1 Metadata completeness - rate

Not applicable.

9.7.2 Metadata - consultations

Not applicable.

10. Cost and Burden

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The questionnaires used for the survey are 100% harmonized with Eurostat's questionnaire. Two forms of questionnaires are used, namely:

- Questionnaire Form 1 and

- Questionnaire Form 2

The questionnaire Form 1 is the questionnaire used for the enterprise and it is used to collect information on the enterprise, such as economic activity, number of persons employed, the form of economic and financial control, number of local units of the enterprise, the main type of employment contract and whether the company belongs to a Group of enterprises. The required time needed to fill in the questionnaire Form 1 ranges from 10 to 70 minutes, depending on the size of the enterprise/service.

The questionnaire Form 2 is used to collect information on the employees. More specifically, it is used to collect information on:

- Individual characteristics of the employee (e.g. sex, age, educational attainment)
- Characteristics pertaining to employment (occupation, position, employment status, total length of service in the enterprise, full or part-time employment)
- Monthly data (the reference month is October of the year when the survey is conducted) on hours paid broken down to overtime pay, on gross earnings broken down to payments related to overtime and to payments related to shift work
- Annual data on gross earnings broken down into payments in kind and payments related to annual bonuses.

It should be noted that the survey is conducted by using the two-stage stratified sampling method, with primary statistical unit the local unit of the enterprise with average annual employment amounting to at least 10 persons and final statistical unit the employee. As a register of local units is not available, the primary unit is considered the enterprise with average annual employment amounting to, at least, 10 persons and classified in sections B-S (excluding O) of Nace Rev.2. Therefore, at a first stage, ELSTAT designs a sample of the enterprises and sets out the number of employees that should be surveyed per enterprise (8 – 250 employees).

At a second stage, during the first visit to the enterprise, a sample of employees is drawn from the payroll list, something that is time-consuming and increases the difficulty level of the survey.

The survey on the structure of earnings is considered by some surveyed enterprises a difficult and time-consuming one. Complaints have been expressed mostly in terms of size of the questionnaire Form 2 and the detailed analysis of required information, in addition to the fact the necessary data must be collected from the local unit and not from the enterprise and finally complaints have been expressed on the size of the sample of employees.

The required time for filling in the questionnaire form 2 ranges from 40 – 60 minutes per employee. The total time needed for filling in the questionnaire form 2 for all the employees of the sample per enterprise depends on the size of the enterprises and the information systems of the enterprise.

The cost of the survey for 2022 amounted approximately to 13,766.34 euros.

11. Confidentiality

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11.1 Confidentiality – policy

The issues concerning the observance of statistical confidentiality by the Hellenic Statistical Authority (ELSTAT) are arranged by articles 7, 8 and 9 of the Law 3832/2010 as in force, by Articles 8, 10 and 11(2) of the Regulation on Statistical Obligations of the agencies of the Hellenic Statistical System and by Articles 10 and 15 of the Regulation on the Operation and Administration of ELSTAT.

Moreover, ELSTAT disseminates the statistics in compliance with the statistical principles of the European Statistics Code of Practice and in particular with the principle of statistical confidentiality. [The confidentiality policy of ELSTAT](#) is available from the link.

11.2 Confidentiality – data treatment

ELSTAT protects and does not disseminate data it has obtained or it has access to, which enable the direct or indirect identification of the statistical units that have provided them by the disclosure of individual information directly received for statistical purposes or indirectly supplied from administrative or other sources. ELSTAT takes all appropriate preventive measures so as to render impossible the identification of individual statistical units by technical or other means that might reasonably be used by a third party. Statistical data that could potentially enable the identification of the statistical unit are disseminated by ELSTAT if and only if:

- a) these data have been treated, as it is specifically set out in the Regulation on Statistical Obligations of the agencies of the Hellenic Statistical System (ELSS), in such a way that their dissemination does not prejudice statistical confidentiality or
- b) the statistical unit has given its consent, without any reservations, for the disclosure of data.

The confidential data that are transmitted by ELSS agencies to ELSTAT are used exclusively for statistical purposes and the only persons who have the right to have access to these data are the personnel engaged in this task and appointed by an act of the President of ELSTAT.

ELSTAT may grant researchers conducting statistical analyses for scientific purposes access to data that enable the indirect identification of the statistical units concerned. The access is granted provided the following conditions are satisfied:

- a) an appropriate request together with a detailed research proposal in conformity with current scientific standards have been submitted;
- b) the research proposal indicates in sufficient detail the set of data to be accessed, the methods of analyzing them, and the time needed for the research;
- c) a contract specifying the conditions for access, the obligations of the researchers, the measures for respecting the confidentiality of statistical data and the sanctions in case of breach of these obligations has been signed by the individual researcher, by his/her institution, or by the organization commissioning the research, as the case may be, and by ELSTAT.

Issues referring to the observance of statistical confidentiality are examined by the Statistical Confidentiality Committee (SCC) operating in ELSTAT. The responsibilities of this Committee are to make recommendations to the President of ELSTAT on:

- the level of detail at which statistical data can be disseminated, so as the identification, either directly or indirectly, of the surveyed statistical unit is not possible;
- the anonymization criteria for the microdata provided to users;
- the granting to researchers access to confidential data for scientific purposes.

12. Comment

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No comments.

Related metadata

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Annexes

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The data of the survey on the structure of earnings are produced at two-digit level of Branches of Economic Activities, by earnings and paid hours, by number of employees, sex, age, educational attainment and average annual earnings.

Estimation process

All local units surveyed were post-stratified by:

- Geographical region (NUTS 1),
- Economic activity (2-digit codes) and
- Employment size class defined as follows:

Size class	Number of employees
1	1-4
2	5-9
3	10-19
4	20-49
5	50-99
6	100-249
7	250-499
8	500-999
9	1000-1999
10	2000-4999
11	5000+

After applying the post-stratification, h represents the stratum formulated after the post-stratification technique. The extrapolation factor of local units was based on the inverse of the selection probabilities of the local units that belong to the enterprises of the sample, as follows:

$$w_{hi} = \frac{N_h}{n_h} \quad (1)$$

Where:

N_h = the estimated number of local units in the post stratum h

n_h = the number of responding local units in the post stratum h

The extrapolation factor of each employee j that was surveyed (employed in local unit i in stratum h), was calculated using the formula:

$$w_{hij} = w_{hi} \times \frac{M_{hi}}{m_{hi}} \quad (2)$$

In each stratum h (stratum= Geography X Economic Activity X Size class), the estimation $\hat{Y}_h$ of the characteristic y of the survey was calculated using the formula:

$$\hat{Y}_h = \sum_{i=1}^{n_h} \sum_{j=1}^{m_{hi}} w_{hij} \cdot y_{hij} \quad (3)$$

where:

n_h : the number of responding local units of the sample in stratum h

m_{hi} : the number of employees that were surveyed in local unit i in stratum h

w_{hij} : the extrapolation factor of the employee j in local unit i in stratum h

y_{hij} : the value of the characteristic y for the employee j in local unit i in stratum h

In the case that the characteristic y is a qualitative variable (e.g. we are interested in the category “full time employees”), then the specific variable takes the following values:

$y_{hij} = 1$, if the surveyed employee belongs to the category of interest

$y_{hij} = 0$, if the surveyed employee does not belong to the category of interest

Estimation of a basic characteristic in country level

The estimation of a characteristic y in country level, Y_{total} (variable y in country level of a two-digit NACE Rev.2 code of economic activity), is calculated by adding up the estimations $\hat{Y}_h$ of every stratum that contains the local units belonging to the specific two-digit NACE Rev.2 code economic activity.

Generally, in order for the estimations of the survey characteristics to be produced at a higher than stratum level, we add up the estimations of the (final) strata, that form the level under survey.

Comparison between SES and LFS

Gross monthly earnings-SES				Net monthly earnings-LFS		
NACE_1D	Total	Men	Women	Total	Men	Women
B	2.183	2.209	1.968	1.039	1.030	1.131
C	1.460	1.558	1.242	1.002	1.030	924
D	2.813	2.955	2.501	1.237	1.239	1.226
E	1.800	1.776	1.880	899	935	813
F	1.403	1.450	1.239	928	935	824
G	1.315	1.536	1.115	918	993	843
H	1.745	1.805	1.595	1.237	1.266	1.091
I	1.086	1.124	1.045	838	867	806
J	1.967	2.203	1.638	1.157	1.192	1.112
K	2.701	3.028	2.423	1.346	1.535	1.162
L	1.760	1.942	1.519	1.294	1.407	921
M	1.783	1.931	1.629	1.064	1.134	998
N	1.099	1.196	1.002	896	927	849
P	1.378	1.430	1.352	1.071	1.197	1.018
Q	1.493	1.689	1.412	1.026	1.157	980
R	1.085	1.161	977	843	908	790
S	1.319	1.395	1.261	906	943	862

Comparison between SES and National Accounts

2022_Sep24

NA_Employees		SES	NA_per head		SES
SECTIONS B TO S	EEM	Number of employees	D.11/EEM	Annual average earnings per employee	
B	8.589	4.690	25.912	27.913	
C	334.116	253.952	18.742	22.993	
D	22.204	18.842	43.037	32.728	
E	26.568	13.457	32.833	20.922	
F	119.101	63.117	11.014	23.391	
G	594.478	383.931	13.594	22.796	
H	200.059	101.462	29.115	21.668	
I	579.527	434.581	5.836	12.062	
J	96.751	70.352	26.548	24.100	
K	66.374	44.876	35.562	35.755	
L	18.856	12.038	11.396	20.995	
M	159.656	94.193	17.963	35.622	
N	140.546	116.965	11.221	13.218	
P	346.707	235.571	15.778	16.497	
Q	239.192	189.531	17.892	19.892	
R	66.446	37.572	9.955	11.774	
S	96.396	45.446	16.080	15.219	
TOTAL		3.115.566	2.120.574	15.555	19.703
B -N	2.366.825	1.612.456	15.422	20.460	

Variable: Gross earnings in the reference month

Estimate	Standard Error	Coefficient of Variation
2.986.166.292,38	27470400,00	,009

Variable: Average Gross Hourly Earnings in the reference month

Ratio Estimate	Standard Error	Coefficient of Variation
9,513	0,059	,006

Variable: Gross earnings in the reference month

Employment Status	Gender	Estimate	Standard Error	Coefficient of Variation
Full-time	M	1.580.714.807,78	19091668,89	,012
	F	1.209.042.410,44	15654176,55	,013
Part-time	M	81.231.221,22	3436921,58	,042
	F	115.177.852,93	3089766,67	,027

Variable: Average Gross Hourly Earnings in the reference month

Employment Status	Gender	Ratio Estimate	Standard Error	Coefficient of Variation
Full-time	M	10,523	0,085	,008
	F	9,007	0,063	,007
Part-time	M	6,452	0,182	,028
	F	6,830	0,089	,013

Variable: Gross earnings in the reference month

NACE Rev.2	Estimate	Standard Error	Coefficient of Variation
B	10.236.013,87	179182,27586	,018
C	370.729.661,64	2551472,08419	,007
D	53.009.868,44	456713,30654	,009
E	24.226.042,18	436682,45807	,018
F	88.535.565,29	2665721,75024	,030
G	505.007.568,67	6259651,26428	,012
H	177.089.478,69	3556340,45744	,020
I	471.891.772,77	15377538,56582	,033
J	138.415.588,82	3533465,03782	,026

K	121.197.129,99	16746866,3307 2	,138
L	21.184.999,85	1382628,61579	,065
M	167.935.966,73	3586596,71075	,021
N	128.528.776,38	2221420,18248	,017
P	324.549.979,94	9636808,21138	,030
Q	282.911.514,08	6485340,46663	,023
R	40.779.905,93	1342686,36237	,033
S	59.936.459,11	1700721,11568	,028

Variable: Average Gross Hourly Earnings in the reference month

NACE Rev.2	Ratio Estimate	Standard Error	Coefficient of Variation
B	13,613	,240	,018
C	9,219	,052	,006
D	17,647	,114	,006
E	11,216	,167	,015
F	8,963	,208	,023
G	8,809	,094	,011
H	10,986	,154	,014
I	8,051	,219	,027
J	12,655	,240	,019
K	16,943	,538	,032
L	11,619	,609	,052
M	11,446	,223	,019
N	7,719	,114	,015
P	10,548	,156	,015
Q	8,820	,117	,013
R	8,199	,206	,025
S	8,828	,151	,017

Variable: Gross earnings in the reference month

ISCO	Estimate	Standard Error	Coefficient of Variation
1	144.782.641,14	7858093,97072	,054
2	821.212.829,76	14024071,9276 7	,017
3	301.652.576,09	10777235,5376 9	,036
4	510.413.229,28	9392080,91824	,018
5	546.230.304,62	8116106,91238	,015
6	12.242.131,05	662816,14393	,054
7	172.820.817,67	2790913,39411	,016
8	198.943.796,19	3251626,44002	,016
9	277.867.966,58	5281785,74295	,019

Variable: Average Gross Hourly Earnings in the reference month

ISCO	Ratio Estimate	Standard Error	Coefficient of Variation
1	21,790	,822	,038
2	12,414	,113	,009
3	10,424	,160	,015
4	9,424	,092	,010
5	7,261	,047	,006
6	7,724	,161	,021
7	9,189	,066	,007
8	8,814	,068	,008
9	6,978	,049	,007

Variable: Gross earnings in the reference month

AGE_BAND	Estimate	Standard Error	Coefficient of Variation
1 under 20	1.819.193,68	290807,75195	,160
2 20-29	249.489.424,22	6593582,95161	,026
3 30-39	605.121.463,35	9997939,34378	,017
4 40-49	955.685.697,74	13422463,18040	,014
5 50-59	908.311.105,22	13278682,55465	,015
6 60+	265.739.408,16	6098749,87278	,023

Variable: Average Gross Hourly Earnings in the reference month

AGE_BAND	Estimate	Standard Error	Coefficient of Variation
1 under 20	5,621	,232	,041
2 20-29	6,550	,076	,012
3 30-39	8,254	,094	,011
4 40-49	9,889	,074	,007
5 50-59	10,874	,082	,008
6 60+	12,072	,197	,016

Variable: Gross earnings in the reference month

NUTS1	Estimate	Standard Error	Coefficient of Variation
1	580.462.970,09	9330213,21070	,016
2	403.986.171,18	6380364,36791	,016
3	1.663.390.869,37	20885991,40104	,013
4	338.326.281,74	13807110,88779	,041

Variable: Average Gross Hourly Earnings in the reference month

NUTS1	Ratio Estimate	Standard Error	Coefficient of Variation
1	8,693	,084	,010
2	8,595	,089	,010
3	10,155	,073	,007
4	9,312	,334	,036

Variable: Gross earnings in the reference month

ISCED	Estimate	Standard Error	Coefficient of Variation
Isced 1	104.409.547,97	3.572.925,56	,034
Isced 2	168.372.152,31	4.187.654,93	,025
Isced 3	1.005.419.259,66	9.937.520,01	,010
Isced 4	303.851.802,62	5.764.247,12	,019
Isced 5	264.635.862,18	4.500.705,72	,017
Isced 6	645.401.502,16	12.242.977,05	,019
Isced 7	433.928.701,98	10.838.942,41	,025
Isced 8	60.147.463,49	6.882.915,08	,114

Variable: Average Gross Hourly Earnings in the reference month

ISCED	Ratio Estimate	Standard Error	Coefficient of Variation
Isced 1	7,66	0,09	,012
Isced 2	7,58	0,09	,011
Isced 3	7,93	0,05	,006
Isced 4	8,84	0,10	,011
Isced 5	9,40	0,09	,009
Isced 6	11,47	0,09	,008
Isced 7	14,85	0,18	,012
Isced 8	17,99	0,43	,024

Variable: Gross earnings in the reference month

Size_class	Estimate	Standard Error	Coefficient of Variation
10_49	601.467.252,55	18.017.432,94	,030
50_249	948.667.345,19	14.195.234,21	,015
250_499	320.011.301,07	12.076.326,51	,038
500_999	309.736.887,96	12.200.618,25	,039
1000+	806.283.505,60	22432073,38	0,03

Variable: Average Gross Hourly Earnings in the reference month

Size_class	Ratio Estimate	Standard Error	Coefficient of Variation
E10_49	8,13	0,17	,021
E50_249	9,40	0,08	,008
E250_499	9,84	0,11	,012
E500_999	10,18	0,17	,016
1000+	10,60	0,13	0,01